

PROJECT REPORT

Project Title: ComicCraft - AI Comic Story Creator using Gemini Models

1. Team Details

- **Team ID:** SWTID-2026-7649
- **Team Size:** 5
- **Team Leader:** N.Sopya
- **Team Members:** S S Sathyabala, Mohamed Ajmal, Mohamed Irfan P, Mohamed jubair.P

2. Abstract

ComicCraft is an AI-driven web application designed to dynamically generate personalized comic book stories and illustrations. Recognizing that creating visual literature requires rare artistic skills, this application leverages Generative AI to tailor comic routines based on an individual's story prompt, setting, tone, and preferred art style. Built on a fast and modern FastAPI framework, ComicCraft provides a seamless user experience from data input to real-time AI generation, layout formatting, and PDF export.

3. Project Objectives

- To develop a responsive web application that collects user story ideas and visual preferences.
- To integrate Google Gemini Generative AI to create structured, panel-by-panel comic storylines and dialogues instantly.
- To integrate Hugging Face Diffusers (Stable Diffusion) for generating high-quality comic illustrations.
- To implement seamless PDF compilation using FPDF, ensuring users can download their generated comics in a structured layout.

4. Technology Stack

- **Programming Language:** Python 3
- **Backend Framework:** FastAPI
- **AI Integration (Text):** Google Gemini API (gemini-1.5-flash & gemini-1.5-pro)
- **AI Integration (Image):** Hugging Face Stable Diffusion
- **PDF Exporter:** FPDF Library
- **Frontend:** Jinja2 Templates, HTML, CSS

- **Server:** Uvicorn

5. System Architecture & Flowchart

The application follows a linear client-server architecture with dual external API integration:

1. **User Input:** The user submits their details (Prompt, Character, Setting, Tone, Style) via the Jinja2 HTML form.
2. **FastAPI Route:** The backend receives the POST request and initiates the AI pipeline.
3. **Gemini API Processing:** A crafted prompt is sent to Google Gemini to generate the 5-panel story outline and dialogue.
4. **Image Generation:** Stable Diffusion API processes the scene descriptions to generate comic-style images.
5. **PDF Compilation:** The backend layout builder compiles the text and images into a structured PDF.
6. **Jinja2 UI:** The final customized comic is rendered elegantly back to the user on the preview page, with a download option.

Data Flow: User UI -> FastAPI (routes.py) -> Gemini API & Stable Diffusion -> Layout Builder & FPDF -> Result UI

6. Key Code Snippets

A. AI Image Generation (image_generator.py)

```
Python
import urllib.request
import json
import os
from PIL import Image
import io

def generate_image(prompt, path):
    hf_token = os.getenv("HF_API_KEY")
    api_url = "https://api-inference.huggingface.co/models/runwayml/stable-
diffusion-v1-5"
    headers = {"Authorization": f"Bearer {hf_token}", "Content-Type":
"application/json"}
```

```
payload = {"inputs": f"{prompt}, high quality comic book art style"}
```

```
req = urllib.request.Request(api_url,
data=json.dumps(payload).encode('utf-8'), headers=headers)
response = urllib.request.urlopen(req, timeout=30)
img = Image.open(io.BytesIO(response.read())).convert('RGB')
img.save(path, format='JPEG', quality=95)
return path
```

B. FastAPI Routing (routes.py)

Python

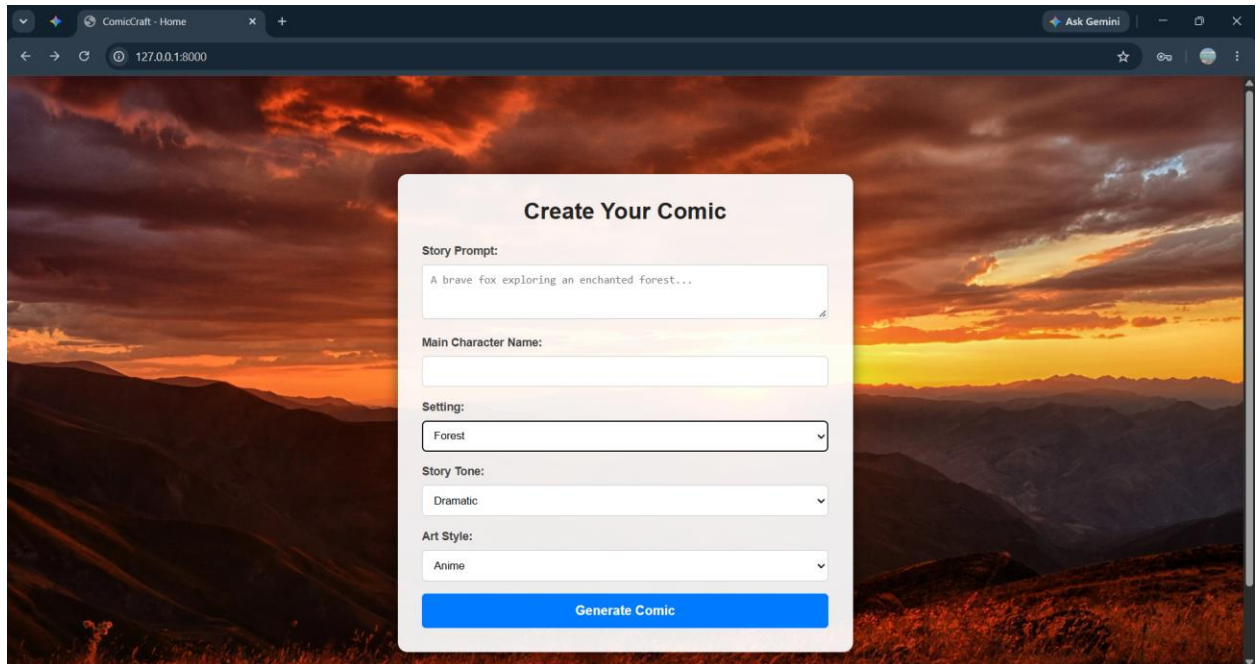
```
@router.post("/generate", response_class=HTMLResponse)
async def generate_comic(request: Request, prompt: str = Form(...),
character_name: str = Form(...)):
    # AI Generation Steps
    full_prompt = f"{prompt}. Character: {character_name}"
    outline = generate_outline(full_prompt)
    full_story = generate_story(outline)
    images = [generate_image(p["image_prompt"], f"panel_{i}.jpg") for
i, p in enumerate(outline)]

    # Layout and PDF Export
    layout = build_comic_layout(images, full_story, outline)
    pdf_path = save_pdf(layout)

    # Render UI
    return templates.TemplateResponse("comic_preview.html",
{"request": request, "layout": layout, "pdf_path": pdf_path})
```

7. Output Screenshots:

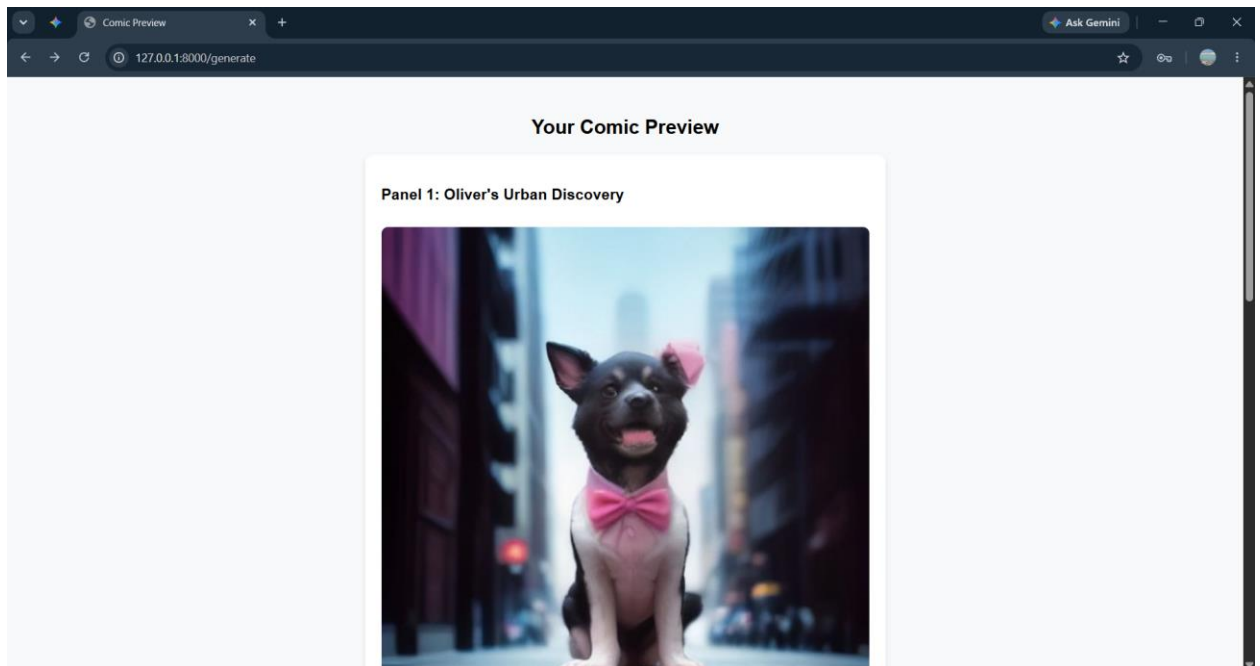
- **Screenshot 1: Homepage (User Input For**



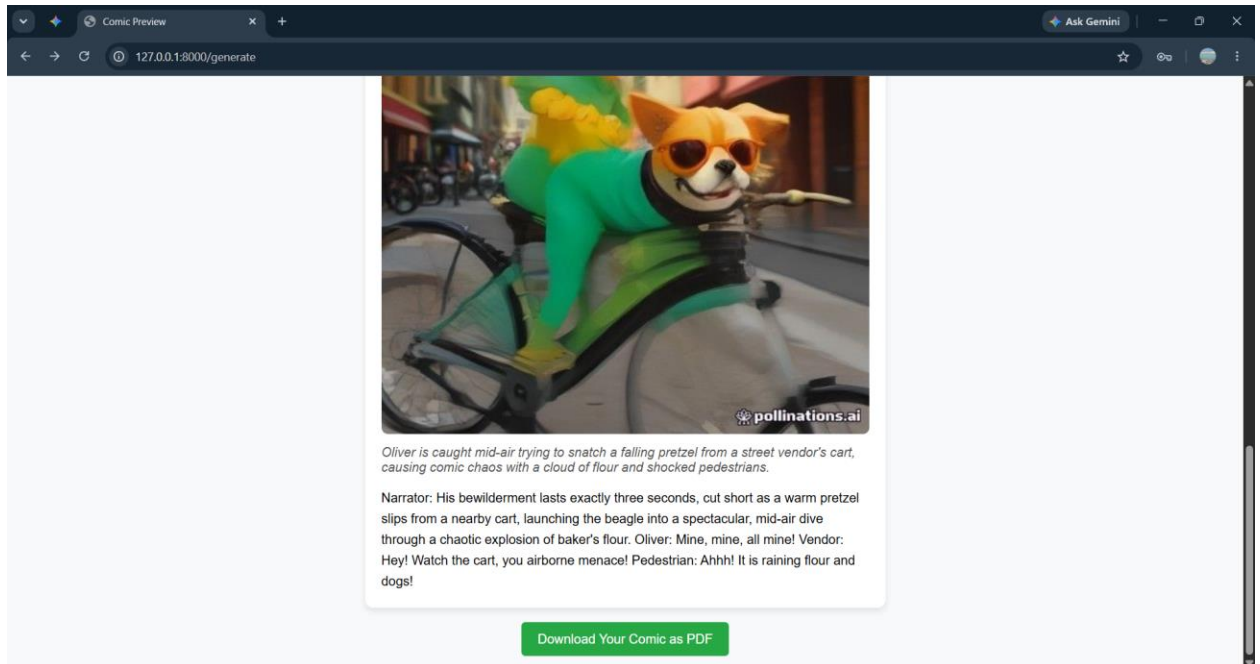
A screenshot of a web browser showing the 'ComicCraft - Home' page. The browser's address bar displays '127.0.0.1:8000'. The page features a background image of a sunset over a mountain range. In the center, there is a white form titled 'Create Your Comic'. The form contains the following fields and options:

- Story Prompt:** A text input field containing 'A brave fox exploring an enchanted forest...'. There is a small icon of a pencil to the right of the field.
- Main Character Name:** An empty text input field.
- Setting:** A dropdown menu with 'Forest' selected.
- Story Tone:** A dropdown menu with 'Dramatic' selected.
- Art Style:** A dropdown menu with 'Anime' selected.
- Generate Comic:** A blue button at the bottom of the form.

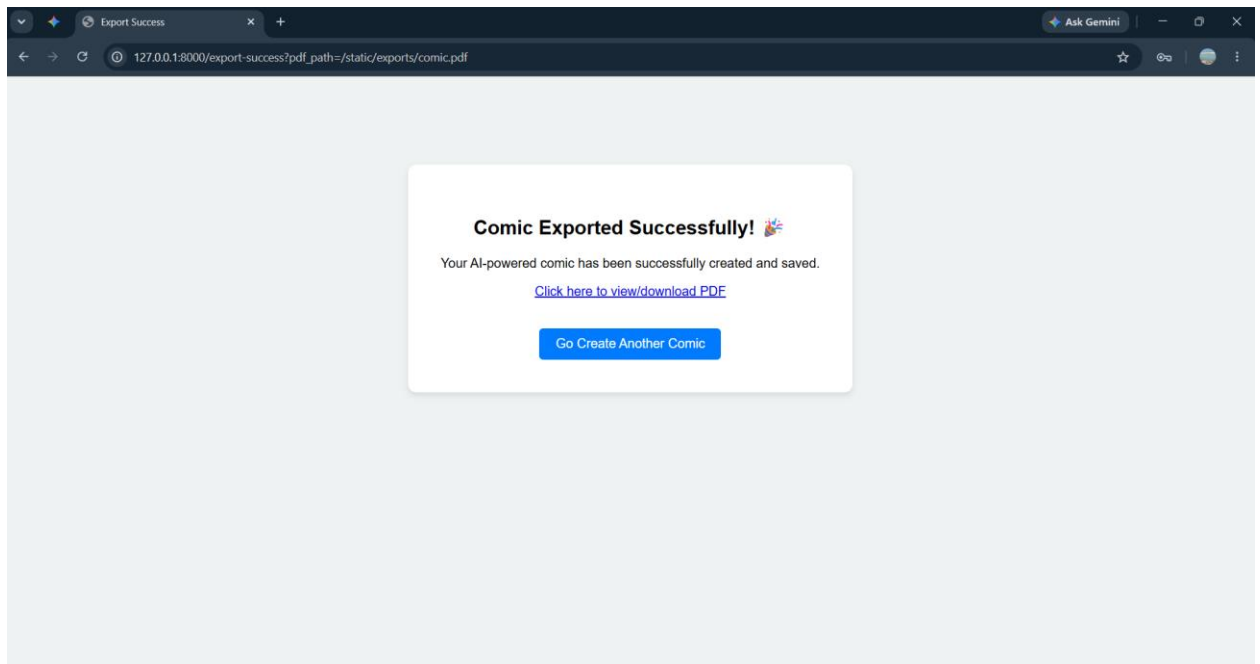
- **Screenshot 2: Loaded (1) (Comic Preview - Panel 1)**



- **Screenshot 3: Bottom of the web (Comic Preview - Continued)**



- **Screenshot 4: Download PDF inside (Export Success / PDF View)**



8. Conclusion:

The ComicCraft project successfully demonstrates the integration of modern web frameworks (FastAPI) with advanced Generative AI (Google Gemini and Stable Diffusion). By automating personalized comic book generation and visual storytelling, the application serves as a scalable prototype for creative tech solutions. The project effectively met all requirements, showcasing robust backend routing, multi-model AI API integration, dynamic layout rendering, and automated PDF formatting, bridging the gap between textual imagination and visual art.